

LOVABLE CLONE

SYSTEM ARCHITECTURE

An AI app-builder where a chat prompt becomes a running, previewable app.
Seven Spring Boot services, choreographed over Kafka and Feign, deployed independently to **GKE Autopilot** behind a TLS-terminating ingress.

CLIENT

ENTRY POINT

Browser

React + Vite SPA

HTTPS

EDGE

NGINX-INGRESS · CERT-MANAGER

TLS Ingress

Let's Encrypt — HTTP-01 for the apex/api, DNS-01 (GoDaddy webhook) for the preview wildcard

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api.lovableclone.in

*.previews.lovableclone.in

GATEWAY

SPRING CLOUD GATEWAY

api-gateway

JWT validation · CORS · routing

NODE · REDIS-BACKED

lovable-me-proxy

Routes preview subdomains to live project pods

CORE SERVICES

AUTH · BILLING

account-service

Signup/login (JWT), Stripe checkout & webhooks, usage limits

Feign (sync)
getUserByEmail

Kafka (async)
UserSignedUpEvent

PROJECTS · FILES · PREVIEWS

workspace-service

Members & invites, file storage, k8s preview orchestration

Kafka saga (async)
FileStoreRequest / Response

AI CHAT · CODEGEN

intelligence-service

Streams LLM output (SSE) that edits project files

DATA LAYER

Postgres + pgvector

Relational data, per-service DBs

account

workspace

intelligence

Redis

Preview hostname routing

workspace

MinIO

Project file storage

workspace

Kafka

Cross-service event bus

account

workspace

intelligence

CONTROL PLANE · USED BY EVERY SERVICE ABOVE

- **config-service** (Spring Cloud Config, pulls from a private git repo) · **discovery-service** (Eureka, local/dev)

— Synchronous (HTTP / Feign) - - - Asynchronous (Kafka event) — Data / infra dependency

- Spring Boot
- Spring Cloud Gateway
- Kafka
- PostgreSQL · pgvector
- Redis
- MinIO
- Kubernetes (GKE Autopilot)
- nginx-ingress
- cert-manager
- Spring AI
- React · Vite

lovableclone.in — an independent, fully self-hosted distributed system.